

Mohamed Bassam Hassan Abdelmegeed

AI Engineer | Agentic AI | Machine Learning | Computer Vision | NLP

Dubai, UAE | UAE Golden Visa Holder | Available for Immediate Joining
(+971) 50 361 3525 | mobassam135@gmail.com | [Portfolio](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI Engineer and Co-Founder with hands-on experience building secure, end-to-end AI products across agentic systems, multimodal applications, computer vision, NLP, and time-series forecasting. Strong in Python, FastAPI, LangGraph, PyTorch, TensorFlow, YOLO11, PostgreSQL/pgvector, and modern web deployment. Built production-oriented platforms with tenant-isolated retrieval, human approval workflows, real-time streaming, audit controls, resilient APIs, and automated testing.

CORE TECHNICAL SKILLS

Languages	Python, TypeScript/JavaScript, SQL
Agentic AI & LLMs	LangGraph, multi-agent orchestration, hybrid RAG, MCP, OpenAI API, Anthropic API, Hugging Face
ML, DL & NLP	PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, LightGBM, BERT, LSTM, Pandas, NumPy
Computer Vision	YOLO11, OpenCV, CUDA
Backend & Data	FastAPI, Flask, REST APIs, WebSockets, Pydantic, PostgreSQL, pgvector, SQLite, BM25/RRF
Frontend, Cloud & DevOps	React, Next.js, Tailwind CSS, Docker, Azure Service Bus, Azure Key Vault, Git/GitHub, Pytest, n8n

PROFESSIONAL EXPERIENCE

Co-Founder & Head of AI | Infeon Jul 2026 - Present
Dubai, UAE & Cairo, Egypt

- Lead the technical architecture and delivery of enterprise AI platforms, multi-agent systems, and multimodal compliance solutions.
- Design real-time vision, language, and voice pipelines that connect LLMs, computer vision, speech synthesis, and production web interfaces.
- Define security, privacy, and reliability standards, including zero-trust API boundaries, protected webhook ingress, sanitised logging, health checks, and failover handling.
- Coordinate FastAPI services, Next.js control centres, cloud integrations, and deployment pipelines from prototype through production readiness.

Machine Learning Intern | CodeAlpha 2024
Remote

- Built and evaluated machine-learning models using SVMs, neural networks, TensorFlow, Keras, and OpenCV for classification and object-recognition tasks.
- Measured model quality using ROC curves, confusion matrices, precision, recall, and related classification metrics.

EDUCATION

Bachelor of Computer Science (Artificial Intelligence) Graduated Jul 2025
Arab Open University (AOU) - Dual Degree Program verified by
The Open University (UK)

SELECTED AI PROJECTS

FLAGSHIP Dark Agent - Governed Multi-Agent AI Mission-Control Platform

Flagship Agentic AI System

Technology: Python, FastAPI, LangGraph, Next.js, TypeScript, PostgreSQL/pgvector, BM25/RRF, WebSockets, Clerk, Stripe, Docker, Azure Service Bus, Azure Key Vault, MCP, OpenAI, Anthropic

- Architected a governed LangGraph workflow with a Planner, parallel specialist agents, a Synthesizer, and an adversarial Red Team validator; added human plan approval and real-time WebSocket streaming.
- Implemented tenant-isolated hybrid retrieval with pgvector semantic search, BM25 keyword ranking, and Reciprocal Rank Fusion, plus episodic memory distillation and client-side PII sanitisation.
- Secured the SaaS control plane with Clerk authentication, idempotent Stripe webhooks, database-enforced model routing, rate limits, deterministic audit logs, and zero-retention Ghost Mode.
- Built production foundations for Docker and Azure, including Service Bus workers, Key Vault-backed read-only MCP connectors, durable queues, provider fallback, and 309 backend tests.

IMPACT Contract Verifier AI - Multimodal Compliance & Verification Engine

Arabic/English document analysis for migrant-worker contract protection

Technology: FastAPI, Next.js, Twilio WhatsApp Webhooks, Google Gemini Vision, ElevenLabs TTS, async processing, health monitoring

- Architected an automated document-verification pipeline that extracts employment terms from mobile contract photos and identifies potentially harmful discrepancies in real time.
- Engineered an image-to-audio workflow that converts structured findings into localized voice alerts delivered through WhatsApp, targeting a sub-10-second response path.
- Improved backend resilience with asynchronous queues, sensitive-data log sanitisation, readiness monitoring, and clear integration boundaries for external services.
- Designed a Next.js control centre for live metrics, system health, integration status, and guided self-service onboarding.

GRADUATION AI Stock Price Prediction - Market Analysis & Paper-Trading Platform

Final-Year Graduation Project

Technology: Python, Flask, TensorFlow/Keras, Scikit-learn, XGBoost, LightGBM, ExtraTrees, Alpaca API, Chart.js, time-series forecasting

- Developed a multi-model forecasting platform combining LSTM, XGBoost, LightGBM, and ExtraTrees to analyse market direction and support paper-trading decisions.
- Integrated the Alpaca Paper Trading API to place simulated orders and synchronise a portfolio dashboard with near-real-time profit-and-loss tracking.
- Built a quantitative risk and market-regime engine with historical backtesting, walk-forward validation, and trade setups containing entry, stop-loss, and take-profit levels.
- Created an explainable dashboard with confidence scoring and engineered indicators including RSI, VWAP, ATR, and MACD; added resilient data-ingestion fallbacks for provider and rate-limit failures.

SELECTED AI PROJECTS - CONTINUED

COMPUTER VISION AI Industrial Safety Monitoring System

Real-Time Computer Vision for PPE and Zone Compliance

Technology: Python, YOLO11, OpenCV, PyTorch, CUDA, n8n, Gmail automation

- Engineered a real-time safety-monitoring system that detects people and Hardhat/No-Hardhat conditions using YOLO11 and OpenCV.
- Implemented Safe Zone and Danger Zone logic so alerts are triggered only when a relevant PPE violation occurs inside a restricted area.
- Added dual-model validation, cooldown logic, incident logging, and automated screenshot capture to reduce false alerts and preserve evidence.
- Connected n8n workflows to send immediate Gmail notifications with incident details and image attachments.
- Optimised inference with CUDA acceleration for responsive desktop deployment.

ADDITIONAL AI APPLICATIONS

AI Study Assistant

NLP Document Summarisation Web Application

Technology: Python, Flask, Hugging Face Transformers, BART, PDF processing

- Built a BART-based summarisation application for direct text input and multi-page PDF documents.
- Developed Flask API endpoints and chunking logic to process long documents without exceeding model memory limits.
- Designed the workflow to clean, segment, summarise, and return readable results through a simple web interface.

AI Writer SaaS

Full-Stack Text Generation and Rewriting Application

Technology: Python, Flask, SQLite, REST APIs, authentication, PythonAnywhere

- Developed and deployed a full-stack AI application for text generation, rewriting, and user-guided content improvement.
- Implemented secure authentication, session management, persistent history, and RESTful generation endpoints using Flask and SQLite.
- Deployed the production application on PythonAnywhere.

ADDITIONAL INFORMATION

Work authorisation: UAE Golden Visa holder; available for immediate joining in the UAE.

Languages: Arabic and English.

Portfolio: mohamed-bassam.vercel.app | github.com/mohamedbassam-1